What spontaneous replicators pass on

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Review copy. Author declarations and release accessions are specified in the corresponding sections at the end. Result-path source brackets resolve from `experiments/`*, unless an absolute path is given; they identify records in the Source Data and provenance index. Numbered citations refer to the reference list. “Core C1” (the corrected core) denotes corrected Z80 semantics; “registered experiment C1” (the displacement study) and “registered experiment C2” (the 6502 retention study) denote the registered capacity studies. Registered outcomes, exploratory analyses, constructions and theorems retain separate status. Figures refer to* `manuscript/figures/out/v8/`*; Supplementary Notes and Tables refer to* `manuscript/SI_v8.md`*.*

Abstract

Darwinian evolution requires reproduction to transmit the differences on which selection acts. Self-copying programs arise spontaneously in machine-code soups, but their survival and spread do not show which parental differences their copying transmits. Using Zilion, an execution substrate for comparing copying mechanisms across processor architectures, we followed introduced differences through descendants. A representative early Z80 copier sustained lineages, yet none of 512 tested variants met the registered carriage criterion [`results/branching/SUPP_TABLE.md`]. For a recurring class of overlapping copiers, we derive a conditional count of freely transmissible byte positions set by an inherited displacement byte; registered Z80 interventions and eligible evolved 6502 copiers matched it [`results/capacity/CAPACITY.md`; `results/evolved6502/RETENTION.md`]. In 8086 populations the predicted emergence route failed and registered copying-law tests remained untested, although exploratory interventions agreed with the rule [`results/s8086/tables/worlds.csv`; `results/s8086/tables/law_denominators.csv`; `results/s8086/exploratory/law_assay_summary.csv`]. Independence from partner information, a strict analogue of one aspect of Woese’s Darwinian threshold, is therefore distinct from transmission of parental variation.

Main text

Sustained evolution by natural selection requires heritable variation in reproductive success [1, 2]. Replication alone does not meet this requirement, because a system can reproduce reliably while passing on few of the differences that arise in it. Von Neumann’s self-reproducing automaton copies a description that it also interprets [3], and the distinction between limited and unlimited heredity asks how many alternatives a reproducing system can carry [2]. Some compositional systems return towards built-in states and fail to preserve selectable variants [4], whereas appropriately organized autocatalytic networks can support limited heredity and adaptation [5]. Woese placed the onset of Darwinian evolution proper at a threshold, where cellular organization became integrated enough for vertical descent to dominate the horizontal exchange of an earlier communal regime [6].

Self-reproduction and heredity have long been studied in cellular automata [7], in evolving machine-code systems such as Tierra and Avida [8, 9] and in Fontana's algorithmic chemistry [10]. Earlier digital experiments also examined spontaneous origins and architecture-dependent evolvability [11, 12].

Artificial chemistries of interacting programs have made the origin of replication itself observable. In task-free program soups, self-replicators arise through mutation and interaction without an imposed fitness function [13, 14]; Cicala and colleagues also studied task-selected conditions, but our comparisons concern their task-free controls [14]. These studies trace where copied bytes come from and have already demonstrated load–push and block-copy architectures, faithfully copied random tails and robustness to repeated cycles of mutation and replication [13, 14].

We ask instead which introduced parental alternatives remain recoverable through successive descendants, and whether an inherited change in the copying operation alters that set. Abundance alone cannot answer this. A successful copier can restore its pattern so effectively that it erases most variants introduced into it, and faithful copying of an unmodified parent does not show which introduced variants would remain. We therefore track introduced differences separately from lineage survival, and derive a conditional count of independently variable positions for a specified family of overlapping copiers. To make such comparisons across processors, we built a substrate that traces copying operations in machines with different instruction sets.

Using this substrate, we show that in soups of Z80 machine code, what reproduction preserves is a measurable property of the copying mechanism. Branching populations of a representative first replicator persist while losing tested variants by the registered carriage criterion. In a recurring family of later copiers, a single geometric property of the copy decides which parental bytes reach descendants. The parameter that sets this geometry is itself inherited, and the chemistry of the environment shifts which designs prevail. Comparing processors then separates how replication begins from what it transmits.

Measuring what reproduction preserves

Zilion separates a processor's instruction semantics from the world in which programs meet (Fig. 1a). An encounter places two tapes of L bytes on a ring of 2L bytes, resets the processor to a declared state, executes a bounded number of execution steps starting in the first tape and writes both tapes back. The soup supplies encounters between lattice neighbours, together with background mutation. It has no fitness function and does not replace unsuccessful programs, so a program reproduces only if its execution writes, into some partner, a tape that reproduces in turn [algocell_exp/machines/; /Users/neo/repos/zilion/docs/ARCHITECTURE.md].

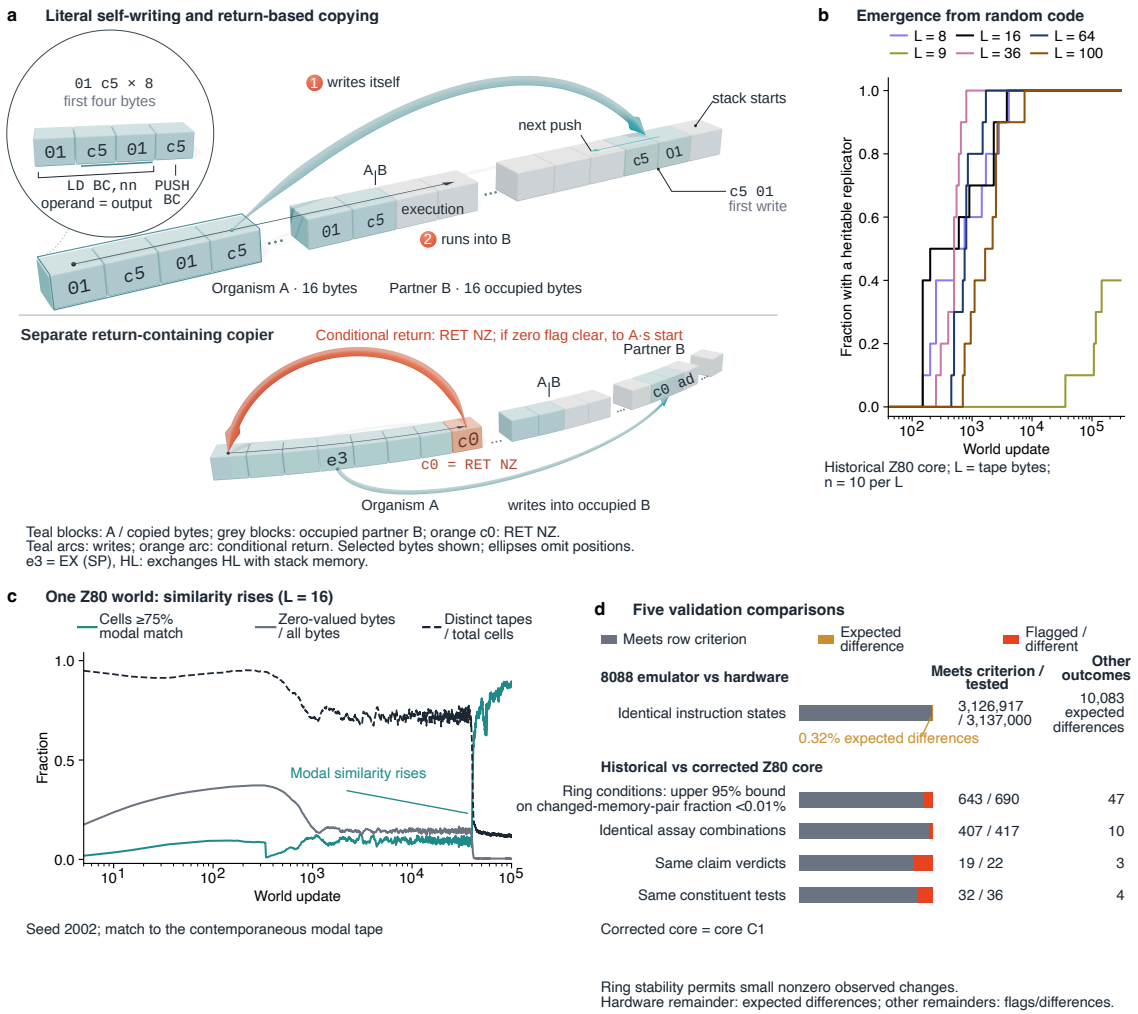


Figure 1. Reproduction made inspectable. a, The schematic shows two Z80 copying mechanisms. The pusher uses the same bytes as instructions and literal output: `01 c5 01` loads BC, and `c5` pushes it into partner memory. Execution can enter the partner; in a separate copier, `RET NZ` returns execution to A's start if the zero flag is clear. These are not a documented ancestor–descendant pair. Selected bytes are shown; ellipses omit positions; `e3` exchanges HL with stack memory. **b,** Historical Stage E detection is one minus Kaplan–Meier survival: six lengths L (bytes), 10 worlds/length, censoring at 300,000 updates. Encounters have a budget of 128 execution steps. Operational heritability requires generation-two normalized culture gain ≥ 0.3. Updates are encounter batches. **c,** One illustrative L = 16 world, seed 2002, contains 20,000 cells. Curves show cells matching the contemporaneous modal tape at $\geq 12/16$ unshifted positions, zero bytes/all bytes and distinct tapes/all cells. Similarity is not a reproduction assay. **d,** Five validation comparisons use distinct denominators: hardware-recorded 8088 instruction states; historical versus corrected Z80 ring conditions; isolated assay combinations; claim verdicts and constituent tests. Ring stability requires a 95% upper bound < 0.01% changed-memory pairs; small nonzero observed changes are permitted. Ochre marks expected hardware divergences; vermilion marks flagged or different comparisons; grey marks comparisons that meet the row-specific criterion. All 10,083 hardware divergences are expected. Queue timing is outside the contract. Corrected core denotes core C1; rows must not be pooled as one accuracy estimate.

A versioned semantic contract specifies everything on which such a claim depends: address reduction on the ring, initial registers and stack phase, the unit of execution and the treatment of undefined instructions. The world supplies cell boundaries, execution time and reset state; the code that exploits them evolves. The distinction matters because an evolved copier can depend on a register value supplied at reset as strongly as on an instruction in its own tape (Methods).

We validated the processor cores against per-instruction reference tests: emulator-generated vectors for the Z80 and 6502 and hardware-recorded vectors for the 8088. The historical populations analysed here were produced with Algocell kernels, and we distinguish their assays from fresh reruns with the corrected Z80 core (core C1) and from current Zilion implementations. The corrected-core reruns retained 19 of 22 claim verdicts and changed 4 of 36 registered components; we report historical failures and corrected-core outcomes side by side [`results/tier2/T2_CLAIMS.csv`; `results/tier2/T2_COMPONENTS.csv`] (Fig. 1d; Methods).

Emergence curves date the culture-based detection of replicators (Fig. 1b), and one illustrative world shows the accompanying changes in sequence similarity (Fig. 1c). Isolated assays then ask what reproduction preserves (Fig. 2), using four measurements that we keep separate throughout. A culture test asks whether a tape's offspring itself reproduces. An execution trace asks whether instruction-stream fetches stay within the replicator's own bytes. Inflow, measured with the parent fixed, is the variation of offspring across partners: how much the partner contributes. Transfer assays follow a deliberately introduced parental difference through repeated reproduction.

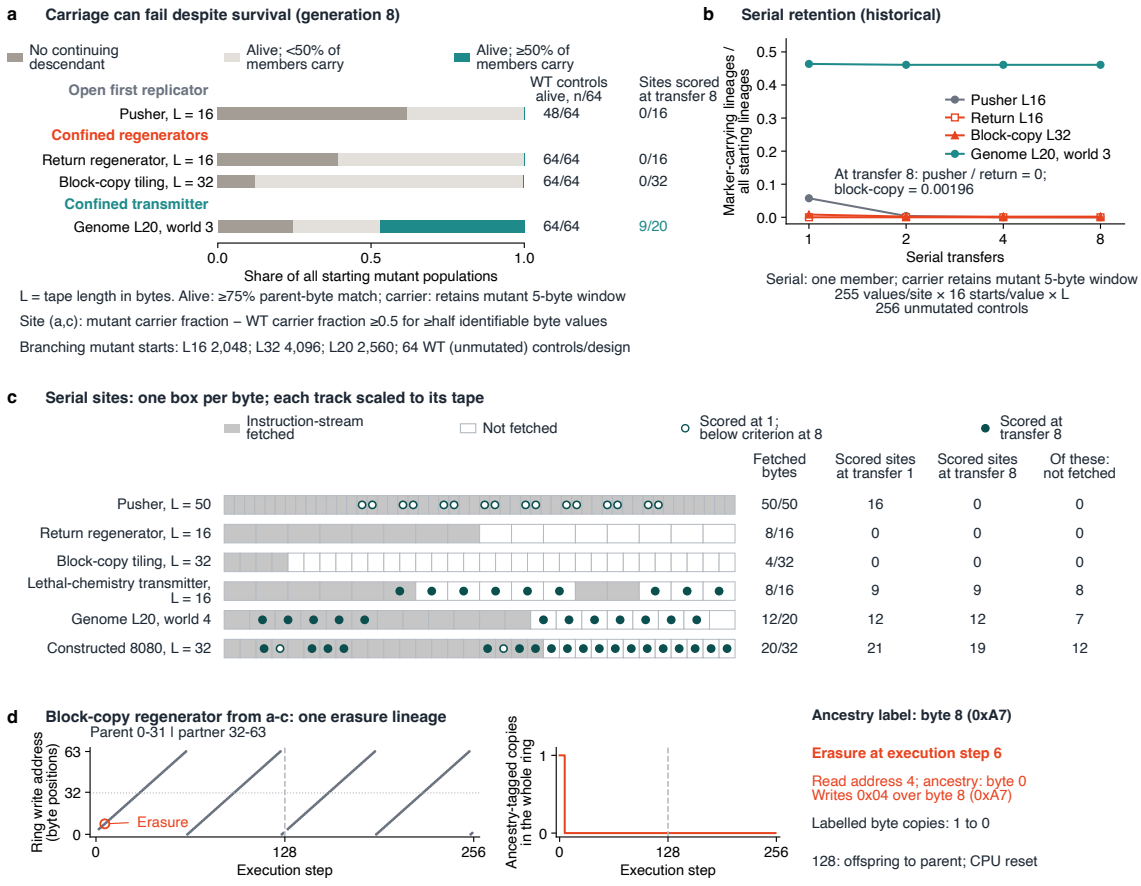


Figure 2. Survival is not retention. a, Registered branching transfers use uniform-byte partners: 32 alternatives/site, four populations/alternative and 64 controls per design. Each member meets two fresh partners; ≥ 75% best-cyclic-shift immediate-parent matches continue, uniformly capped at 32 members. Generation-eight bars partition mutant starts: lost, or alive with < 50% or ≥ 50% of members carrying the marker window. Minority carriers can remain. Markers are centred five-byte windows absent from wild type. Sites score when mutant-minus-control carrier fractions among surviving populations are ≥ 0.5 for at least half the identifiable alternatives. Columns give control survival and site scores. Historical/corrected-core endpoints coincide. **b,** Historical serial assays use these designs: 255 alternatives/site, 16 starts/alternative, eight transfers and 256 controls. Fractions use all mutant starts. **c,** Site maps show one byte/box; tracks scale separately. Grey/white mark instruction-stream fetch/no fetch across

64 reference encounters, including operands. Open circles score at transfer one but fall below the site criterion at transfer eight; filled circles score at transfer eight. The lethal-chemistry L16 tape is a confined transmitter. The L50 pusher uses 32 alternatives/site; others use 255. L20 uses seed 6004 (6003 in a,b). Serial scoring applies this excess criterion to surviving lineages. **d,** One exploratory lineage meets all-zero partners in two encounters of 128 execution steps. At zero-based execution step 6, address 4 supplies original-byte-0 ancestry, overwriting 0xA7 at address 8 and erasing its last ancestry label. At execution step 128, offspring replaces parent and the processor resets. This trace differs from window carriage.

These properties dissociate in both directions. A program can read a partner's bytes without executing them, and it can execute partner addresses after overwriting their contents, so none of the measurements stands in for another (Methods).

Carriage can fail despite survival

In all 80 historical worlds of the registered benign-chemistry cohort, the first detected replicator was open in the sampled encounters: its execution ran on into the partner it was overwriting [results/inflow_insoup/INFLOW_INSOUP.md, X2-2]. Its most common form tiles the tape with `01 c5` [results/stageG/stageG/stage_g_runs.csv], a load–push motif reported for the 8080 by Agüera y Arcas and colleagues and subsequently assayed in the Z80 by Cicala and colleagues [13, 14]. The tile decodes as `01 c5 01`, the instruction `LD BC,$01C5`, followed by `c5`, `PUSH BC`, so the same byte values serve both as opcodes and as the literal data pushed into the partner's memory.

To ask what such a replicator transmits, we carried out a registered branching-transfer experiment (X1), in which every surviving member of a population reproduced against two fresh partners per generation, so that loss of a mutation could be distinguished from extinction of its lineage (Fig. 2a). Under uniform random partners, 48 of 64 unmutated populations of the 16-byte pusher survived for eight generations. Among mutant populations, 38% were still alive at generation eight, yet for none of the 512 tested mutation types did a population meet the registered criterion for carrying its mutation. The registered prediction that the open replicator would retain heritable variation (X1-2) therefore failed, and we report it as failed [results/branching/branching_summary.csv, pusher16, arm U; results/branching/SUPP_TABLE.md; REVISION_PREREG.md, X1]. Failure of this majority-carriage criterion does not establish that every copy of a marker was lost; it shows that the pusher rebuilds its form while losing these variants by the registered criterion.

Three confined Z80 replicators gave contrasting results under the same protocol (Fig. 2a). Two regenerators, one returning control to itself and one using a block copy, preserved every unmutated population and scored no transmissible site, whereas an evolved block-copy transmitter carried nine scored sites through all eight generations [results/branching/branching_summary.csv, ret16, ldir32, genome20_s6003]. A site scores when at least half of the tested identifiable alternative values at that position meet the registered retention criterion. This standard is demanding, and rare variants can persist below it: the 64-byte pusher retained 5 of 2,048 tested mutant types with uniform partners, and the block-copy regenerator retained a few working core variants [results/branching/SUPP_TABLE.md].

The contrast is therefore not between all and nothing but between copiers that preserve broad classes of alternatives at specific positions and copiers that preserve almost none. The confined regenerators and the transmitter differ in what reproduction preserves despite identical survival of their unmutated populations. Historical serial transfers and byte maps show the corresponding retention contrast (Fig. 2b,c), and an ordered trace follows an introduced byte through its overwrite

and the descendant's next reproduction (Fig. 2d). We next ask what, in the copying operation, decides which differences survive.

Overlap sets what is inherited

For the most tractable family of evolved copiers, the answer is geometric (Fig. 3a). Consider a forward copy that begins at the parent's first byte and writes each byte a fixed displacement d ahead of where it was read, where d divides the tape length L, the ordered writes cover the ring and no other write interferes. The first d reads obtain original parental bytes. Thereafter the read pointer reaches bytes that have already been written, so it copies copies: the first d bytes propagate around the whole ring, and every byte beyond them is overwritten before it can be read. A small displacement therefore regenerates a short pattern regardless of what else the parent held, whereas a displacement spanning the whole parent reproduces the entire parental tape.

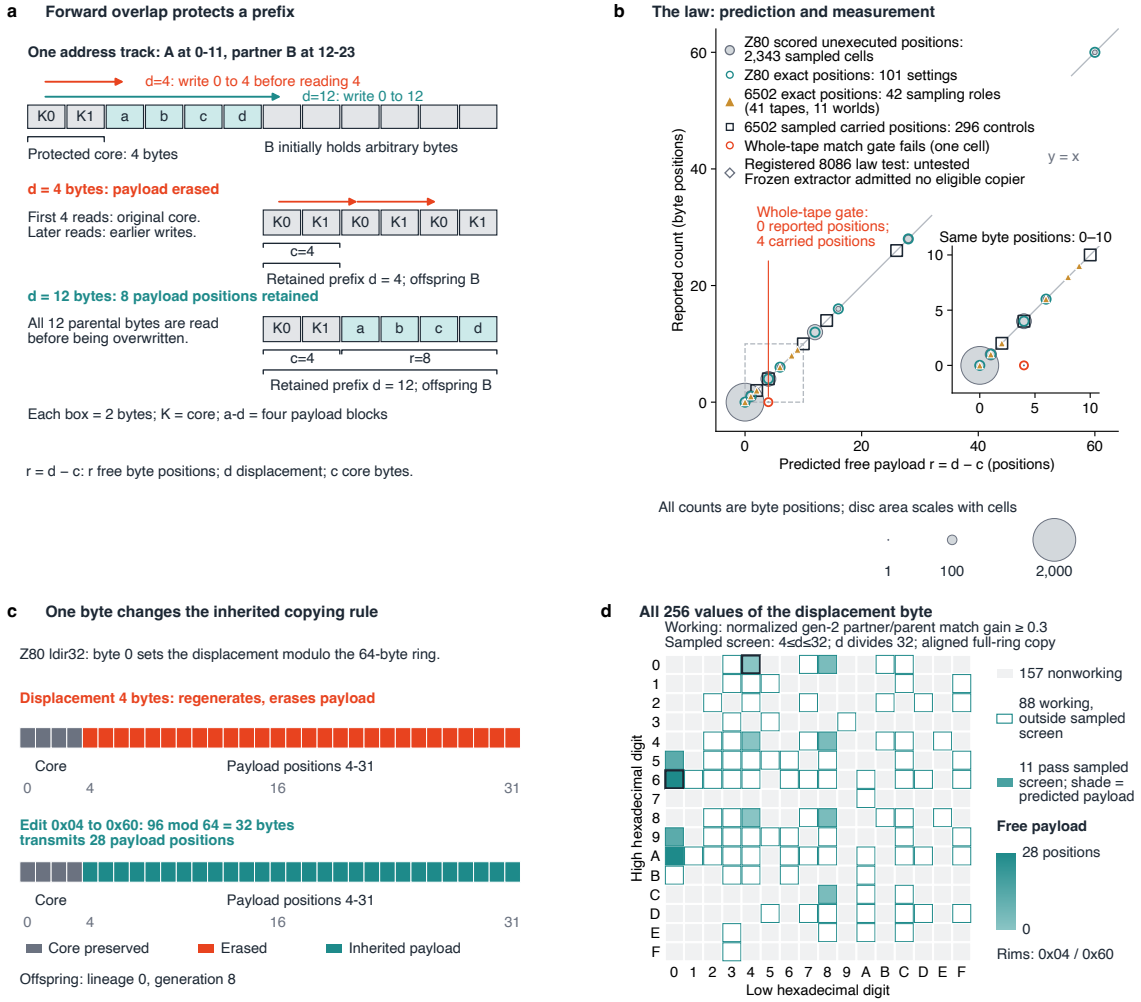


Figure 3. Inherited displacement controls free payload. a, Forward overlap copies two L = 12 tapes; boxes represent two bytes. For fixed core length c and displacement d, predicted free payload r = d − c. Displacement d = 4 regenerates the four-byte core; d = 12 preserves eight payload positions. Initialization, d dividing L, coverage, noninterference and alignment are required. **b,** Predicted versus reported positions (inset: 0–10). Grey discs show sampled unexecuted-site scores for 2,343 exploratory Z80 cells; area denotes count. Teal rings: exact capacities for 101 registered settings (22 original-value controls included; 22 parents). Ochre triangles: exact capacities for 42 eligible 6502 roles (41 tapes, 11 worlds). Squares: 296 constructed 6502 sampled-carriage cases, not exact capacities. Symbols overlap. The red point carries four

positions in targeted transfers but scores zero under the $\geq 75\%$ whole-tape gate. Cells and roles are nested observations. Registered 8086 L1/L2 remain untested; L2 support is suspended. Post hoc assays are not plotted. **c,** Parallel generation-eight descendants use lineage zero. Editing byte 0 (0x04→0x60) gives displacement 96 mod 64 = 32, transmitting 28 payload positions. Grey denotes fixed core, red lost ancestry and teal inherited payload. This intervention is not a spontaneous transition. **d,** An exploratory 256-value scan finds 157 nonworking, 88 working outside the sampled screen and 11 that pass the sampled screen, shaded by predicted payload. Working requires generation-two normalized culture gain ≥ 0.3 with 16 uniform partners. Rims mark 0x04/0x60; eligibility and working scores do not certify universal serial inheritance.

With the copying core of c bytes held fixed inside the first d, and under the stated initialization, offspring alignment and forward coverage, the number of independently free payload positions is $r = d - c$ [theorem: `manuscript/SI_v8.md`, Supplementary Note 1; Methods]. The proof traces every byte of the final ring back to its source and holds for jointly arbitrary payload values. It counts free payload only; working variants of the core itself are additional possibilities that it does not count.

The Z80 soups evolved copiers with this geometry. In the four-byte core `XX 5e ed b0`, the first byte executes as a harmless instruction and is then read as data by `LD E,(HL)` to set the displacement, and `LDIR` performs the overlapping copy. The variant `1e NN ed b0` carries the displacement as an immediate operand. Cicala and colleagues described the four-byte immediate-displacement copier and offset-reading block-copy architectures [14]; here we derive and test the conditional payload count $r = d - 4$ [theorem: `manuscript/SI_v8.md`, Supplementary Note 1]. The core occurs in 2,347 of 15,655 sampled confined heritable cells, so it is a recurring architecture rather than the whole ecology [`results/offset/CORE_FAMILIES.md`].

In an exploratory regrouping of the cells that carry this motif with a displacement dividing the tape length (Fig. 3b), sampled site scores closely match the prediction. Of 2,343 such cells, 2,342 had exactly $d - 4$ scored unexecuted transmissible positions. In all 53 cells whose total score exceeded $d - 4$, the additional sites lay at positions fetched as part of the instruction stream. The single deficit preserves its predicted payload in targeted transfers but fails the whole-tape similarity gate of the population assay [exploratory: `results/crystal/cells.csv`, `results/crystal/cell_exceptions.csv`, `results/crystal/offset_examples.csv`; `results/offset/offset_cells.csv`]. Motif and divisibility alone do not certify every hypothesis of the theorem, and these are sampled scores, not exhaustive capacities (Methods).

The law also corrects a tempting intuition: what matters is the displacement of the copy, not the visible period of the tape. One evolved tape repeats every four bytes, all of them core, yet has six scored transmissible sites, four of them unexecuted payload positions, because its copy displacement is eight [exploratory: `results/crystal/cell_exceptions.csv`, K, L = 32, seed 5006]. Hereditary capacity belongs to the family of sequences that a copying operation preserves, not to the appearance of any one member.

The same operation separates two properties that are easily conflated. Under the theorem's hypotheses, every byte of the offspring descends from the parent's protected prefix, whatever the partner initially held. Such an overlapping regenerator can therefore exclude partner information entirely while erasing variation outside its protected prefix. Closure removes an external source of difference; transmission preserves an internal one.

The copying rule is inherited

If the displacement is encoded in the replicator's own bytes, it should be copied with the core, and a change in it should be inherited. Archived serial-transfer interventions on an evolved regenerator, ldir32, show that it is: when its displacement byte was changed from `0x04` to `0x08` or to `0x60`, the mutant markers persisted through eight transfers [`results/serial_retention/sr_mutants.csv.gz`, ldir32, gen 8, pos 0, val 8 and 96]. Under the theorem these changes predict more free payload, and the Z80 displacement study (registered experiment C1) measured that capacity directly (Fig. 3b,c).

From 22 archived parents we made 101 frozen displacement settings, including original-value controls, with forward and reciprocal passages. Eligibility and predicted coordinates were fixed from code and traces before retention outcomes were observed. In 16 lineages per setting and direction, followed for eight generations, the edited displacement and the whole core were inherited. Every tested alternative at each predicted free coordinate in actual generation-eight descendants was transmitted, sampled joint assignments were transmitted, and erased-coordinate controls lost their original ancestry. Across 1,067,608 direct cases there was no mismatch, with fixed-core capacities from 0 to 60 positions [registered: `results/capacity/edit_eligibility.csv`; `results/capacity/serial.csv.gz`; `results/capacity/capacity_cases.csv.gz`; `results/capacity/CAPACITY.md`]. An inherited change to a copier with no free payload can thus permit most of its tape to vary, and a reciprocal edit restores the original erasure.

An exploratory scan of all 256 values of this byte shows how wide the dial is (Fig. 3d) [exploratory: `results/crystal/alleles_ldir32.csv`]. Ninety-nine values give a working replicator, and 50 of these match the edited value at the aligned position in at least half of first-generation offspring; such equality alone does not certify serial inheritance. The 11 values that pass the sampled eligibility checks span displacements of 4 to 32 bytes and predicted payloads of 0 to 28 positions. The dial is specific to this family: in the same exploratory scan, an evolved transmitter that copies backwards with `LDDR` had only 15 working values, and it lies outside the forward theorem [`results/crystal/alleles_genome20_s6003.csv`].

This is an evolved instance of an old architectural idea. Von Neumann's automaton requires a description that is both used and copied [3]; the displacement byte is used to construct the copy and is copied along with it. Within this copying family it is a heritable trait whose predicted phenotype is the capacity to transmit arbitrary payload. We do not claim that selection favours it for that reason: whether a larger payload benefits a lineage depends on the environment.

Environments sort hereditary designs

The competitions below use two frozen, census-defined byte classes, R and T, corresponding to partial-copy (d < L) and full-tape (d = L) settings; these differ from the site-score classes used in random-population assays (Methods). Without mutation, final shares in mixtures varied widely between worlds, and the registered E1 stability criterion failed: only 4 of 10 corrected-core benign worlds at each length ended in the 0.3–0.7 interval, against the required 7 of 10. This pattern is consistent with drift but does not establish neutrality [`results/tier2/scored/E/REPORT.md`, mutation off; `results/tier2/T2_COMPONENTS.csv`, E1]. The registered mutation-on prediction E2 also failed: the T share was to fall below its start and end between 0.05 and 0.4, but did so in only 2 of 10 benign worlds at L = 16, 0 of 10 lethal worlds at L = 16 and 5 of 10 benign worlds at L = 32, against the required 7 of 10. At L = 16 under benign chemistry, core-carrying cells fell to a median of 5% of all cells, so the share is not interpretable there [`results/tier2/T2_COMPONENTS.csv`, E2;

`results/tier2/scored/E/REPORT.md`, L = 16, mutation on]. With mutation, the chemistry of the zero byte shifts their relative abundance. In the corrected-core reruns of the registered competitions at tape length 32, started with equal shares, median full-tape-class shares, T/(R + T), were 0.05 under benign chemistry, where a zero opcode executes harmlessly, and 0.76 under lethal chemistry, where fetching a zero opcode halts execution. All ten lethal worlds exceeded one half, meeting the registered endpoint [`results/tier2/scored/E/REPORT.md`, L = 32, mutation on; `results/tier2/scored/E/offset_switch.csv`; `results/tier2/T2_COMPONENTS.csv`, E-L32L] (Fig. 4a,b).

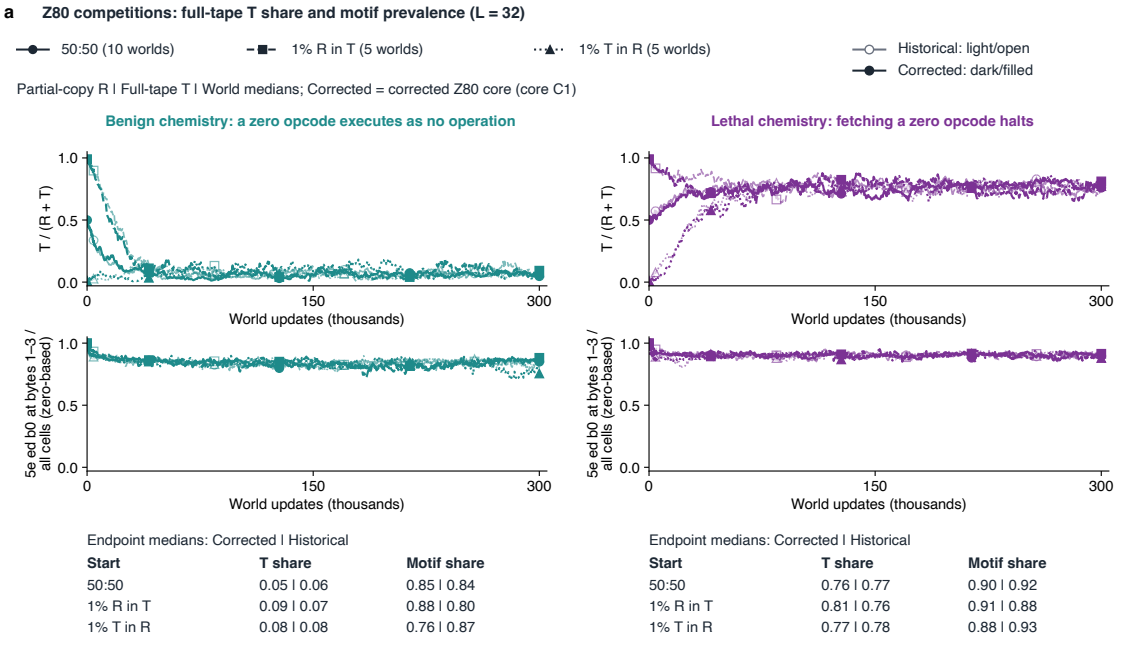


Figure 4. The environment sorts copying designs. a, L32 mutation-on competitions run through 300,000 updates. Frozen census-defined R/T classes denote partial-copy (d < L) and full-tape (d = L) settings, distinct from c's site-score classes. Upper curves show T/(R+T); lower curves show cells carrying `5e ed b0` at bytes 1–3 divided by all 20,000 cells, including other byte-0 variants. Curves are world medians. Each chemistry/implementation has 10 equal-start worlds and five per reciprocal 1% invasion. Styles/markers identify starting mixtures; light/open denotes historical implementations and dark/filled the corrected core (core C1). Under benign chemistry, a zero opcode executes as no operation; under lethal chemistry, fetching a zero opcode halts. Endpoint columns give corrected then historical medians. **b,** Endpoints show relative T share, motif prevalence and distinct T-tail sequences at bytes 4–31. Shapes follow a; horizontal offsets separate points. One corrected-core benign world has R+T = 0: 19 ratios remain overall, four in its five-world T-into-R subgroup. Zero absolute abundance and tail diversity remain included. **c,** The random-code lethality dial uses L16 and 10 worlds per probability/implementation. p denotes halting

probability upon fetching a zero opcode. Open requires partner instruction-stream fetch in $\geq$ 8/16 traces; confinement requires none. Transmitter majority means > 50% of heritable cells are confined with $\geq$ 5 byte sites retaining introduced values under the carriage-excess criterion. Counts retain all worlds. Confinement times are Kaplan–Meier medians (thousands of updates); nondetections are censored at 300,000. Historical DZ2/DZ3 failures remain distinct from corrected-core passes. Endpoints establish neither equilibrium nor its mechanism.

Both invasion directions ended at similar environment-specific medians, with five worlds per direction: 0.09 and 0.08 under benign chemistry and 0.81 and 0.77 under lethal chemistry. The benign T-into-R median of 0.08 uses four defined ratios because one world has R + T = 0; that world retains zero absolute abundance and tail diversity in Fig. 4b [manuscript/figures/out/v8/data/competition_endpoints.csv]. These finite-horizon endpoints are consistent with convergence from either rare starting design but do not establish equilibrium [results/tier2/scored/E/REPORT.md, L = 32]. In the corrected-core lethality dial, transmitters formed a majority of heritable cells in 8 of 10 fully lethal worlds, against 6 of 10 historically, when this registered endpoint failed [results/tier2/T2_DZ.csv; REVISION_PREREG.md, DZ3].

The lethality dial also changes observed route frequencies and confinement times (Fig. 4c), although it does not impose a fixed sequence. With the corrected core, median times to first confinement rose with lethality apart from a single reversal, meeting a registered timing criterion that the historical run had failed with two reversals; the registered endpoint is biased against confinement at high lethality (Methods) [results/tier2/T2.md, DZ2; REVISION_PREREG.md, DZ2]. In an exploratory reanalysis of historical benign worlds, regenerators were the final majority of heritable cells in 17 of 20 worlds at L = 16 and 14 of 20 at L = 20, whereas at L = 50 open cells remained the majority in all 20 worlds. Among the 42 benign worlds containing both classes, regenerators were detected first in 30, an order of detection rather than of ancestry [exploratory: results/ladder/population_worlds.csv; results/ladder/LADDER.md]. Under lethal chemistry the picture differs. In an exploratory assay of the separate historical lethal-chemistry cohort, all ten first detected tapes were confined, had zero sampled partner inflow and carried at most four scored sites, yet by the final sample transmitters were a majority of heritable cells in nine of the ten worlds [exploratory: results/crystal/first_cells.csv, Z80 I; results/ladder/population_worlds.csv, stage I].

Why lethal chemistry favours transmitters remains open. Registered tests failed to explain the environmental balance by the specified toxin effect, by backup cores or by isolated mutational flows; halting payload bytes provided a small protective effect, below the registered explanatory threshold [results/toxin/CAUSAL.md; results/n2/BACKUP.md; results/offset/OFFSET_FLOW.md; REVISION_PREREG.md, N1, N2 and E-mech]. Exploratory encounter assays locate an asymmetry in how the two designs fare against core-free intruders, but they do not yet explain the balance [exploratory: results/offset/INTRUDER.md]. Because a tape is both a copier and material that other programs execute and overwrite, what a replicator preserves changes what its neighbours encounter. A mechanism for the balance will have to predict how the balance moves after an intervention.

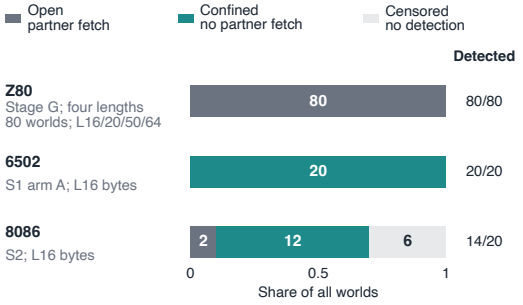
Available copiers need not emerge

The comparison across processors separates an available copying mechanism from the route that evolution takes (Fig. 5a–c). The 6502's accumulator load–push channel is one byte wide. A retrospectively derived theorem excludes an infinite periodic self-writer in this single-register grammar, but not every finite encounter or every instruction family [theorem: results/literal_channel/CHECK.md]. Before the soups were run, a validated census found no nonzero exact

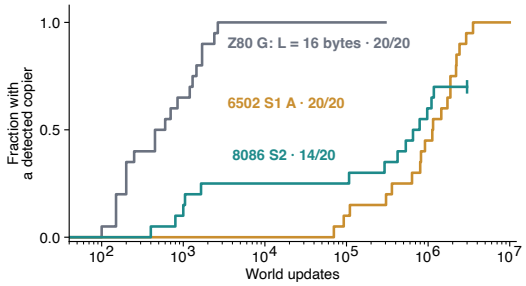
zero-partner self-writer of period one to four at L = 16 under the declared 128- and 512-step arms [results/census6502/table_main.csv]. In registered S1 primary arm A, all 20 first detected replicators were confined, with a median detection time of 1,125,000 updates. Only 14 met the additional registered loop criterion, so prediction P1 was not supported; the ring-position explanation of the remaining six is post hoc [results/s6502/v2/arm_comparison.csv; results/s6502/CORRECTION.md].

S1/S2: open ≥8/16 partner-fetch encounters; confined 0/16; no first cases in 1–7/16

a First-replicator route



b Arrival: first detected heritable copier



Observation ends: Z80 300,000; 6502 10,000,000; 8086 3,000,000 updates (6/20 undetected; censor tick).

c Literal-channel expectation and observed beginnings

Machine	Literal width (bytes / push)	Exact-pusher candidate present (opcode motif)	First route among detected worlds
Z80	2	Yes · 01 c5	Open · 80/80
6502	1	No	Confined · 20/20
8086	2	Yes · B9 51	**12/14 confined; 2/14 open**

Exact pusher: candidate writes its literal instruction bytes into an all-zero partner; routes: evolved worlds

d 8086 convergence: the 5F A4 x y first-copier family (10/14; all ten confined)

World seed	5F A4 = POP DI; MOVSB								x, y: variable bytes								First detection (world updates)	Frozen full-address class (16/16 encounters)
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15		
8086001	5F	A4	63	3D	5F	A4	63	3D	5F	A4	63	3D	5F	A4	63	3D	652,500	One-pass class
8086005	5F	A4	73	5D	5F	A4	73	5D	5F	A4	73	5D	5F	A4	73	5D	975,000	One-pass class
8086006	5F	A4	6D	7D	5F	A4	6D	7D	5F	A4	6D	7D	5F	A4	6D	7D	1,650	One-pass class
8086009	5F	A4	61	3D	5F	A4	61	3D	5F	A4	61	3D	5F	A4	61	3D	292,500	One-pass class
8086012	5F	A4	E2	7D	5F	A4	E2	7D	5F	A4	E2	7D	5F	A4	E2	7D	1,100,000	One-pass class
8086013	5F	A4	7D	FD	5F	A4	7D	FD	5F	A4	7D	FD	5F	A4	7D	FD	1,175,000	Loop class
8086014	5F	A4	E9	3C	5F	A4	E9	3C	5F	A4	E9	3C	5F	A4	E9	3C	532,500	One-pass class
8086017	5F	A4	61	BD	5F	A4	61	BD	5F	A4	61	BD	5F	A4	61	BD	422,500	One-pass class
8086018	5F	A4	69	DD	5F	A4	69	DD	5F	A4	69	DD	5F	A4	69	DD	787,500	One-pass class
8086020	5F	A4	63	FD	5F	A4	63	FD	5F	A4	63	FD	5F	A4	63	FD	107,500	Loop class

Frozen class uses full instruction-start addresses: One-pass = no repeated start; Loop = repeated start.
Labels agree in all 16/16 encounters per tape. Post hoc: all ten repeatedly copy mapped ring positions at displacement d = 4 bytes.
G = historical random-code cohort; S1 A = 6502 primary arm; S2 = coherent-fetch 8086 arm.

Figure 5. Different machines, different beginnings. a, First-copier routes cover all worlds. Historical Z80 Stage G: 80 worlds (20 each at L = 16,20,50,64), all open first. Primary 6502 S1 A: 20 L16 worlds, all confined first. Coherent-fetch 8086 S2: 20 L16 worlds—two open, 12 confined, six censored at 3,000,000 updates. S1/S2: open = partner instruction-stream fetch in $\geq$ 8/16 traces; confined = none. Historical Z80 classification used control-flow and partner-copy evidence. Confinement does not establish partner-content independence. S1's additional loop requirement holds in 14/20, leaving P1 unsupported; S2 passes detection sufficiency but fails the open-route prediction and triggers its kill rule. **b,** Kaplan–Meier detection: 20 L16 worlds/machine; budget of 128 execution steps. Horizons are 300,000 Z80, 10,000,000 6502 and 3,000,000 8086 updates. One terminal mark: six coincident 8086 censors. Events are scheduled detections, not births; protocols differ and timing comparisons are descriptive. **c,** Bytes/push and exact pusher availability follow declared census/reset contracts. The 6502 absence concerns the tested periodic census, not all possible copiers. Availability does not determine establishment. **d,** All ten `5F A4 x y` first-copier tapes (14 detected worlds) appear in physical phase with detection times. `5F A4` encodes `POP DI; MOVSB`. Tile outlines group repeats. Frozen full-address classes hold in 16/16 encounters: eight one-pass and two loop

classifications. The classes use full instruction-start addresses; post hoc traces show all ten tapes repeatedly copying mapped ring positions at displacement d = 4 bytes. Colours identify shared bytes, not certified payload. Exploratory overlap analysis leaves registered verdicts unchanged.

The copying law nevertheless transferred. Constructed 6502 controls gave the predicted sampled carriage in 296 qualifying cases, which are distinct from exact capacities. The complete panel of the 6502 retention study (registered experiment C2) sampled 942 roles from all 80 S1 worlds and, by frozen code-and-provenance rules, admitted 42 sampling roles representing 41 distinct tapes from 11 worlds. Their constrained footprints spanned 6–10 bytes and predicted payloads 0–9 positions. All tested predictions held through generation eight, with no mismatch in direct or serial assays (Fig. 3b; Methods) [construction: `results/census6502/table_closers_law.csv`; registered: `results/evolved6502/eligibility.csv`; `results/evolved6502/geometry.csv`; `results/evolved6502/law_verdict.csv`; `results/evolved6502/exact_cases.csv.gz`; `results/evolved6502/exact_serial.csv.gz`]. Roles and interventions are nested observations, not independent worlds.

The coherent-fetch 8086 supplied the prospective challenge. Its exact pusher was available: under the frozen reset, the tile `B9 51` decodes as `MOV CX,$B951; PUSH CX` [`results/census8086/table_hits.csv`; `algocell_exp/machines/worlds_8086.json`]. Yet its first detected replicators were predominantly confined (Fig. 5a,c). Registered study S2 found copiers in 14 of 20 worlds, meeting R0, but only 2 of those 14 began open, failing R1; the 12 confined beginnings triggered the route kill rule, and the census-motivated open-beginning claim is withdrawn. Detection had a Kaplan–Meier median of 652,500 updates, with six worlds censored at the fixed horizon (Fig. 5b); the timing comparison is descriptive [registered: `results/s8086/tables/worlds.csv`; `results/s8086/tables/kaplan_meier.csv`; `results/s8086/S2.md`]. Availability had supplied a possible route, not its probability of arrival or establishment.

The registered law tests did not resolve the question. L1 found no eligible evolved member of its narrow REP family and is untested. Amended L2 also admitted none: 724 certificates remained unresolved, 708 of them at the unmodelled `POP ES`. L2 is therefore untested with support suspended; it is neither a law failure nor a pass with no mismatches. The R2 measures of mechanism occurrence and persistence were demoted to descriptive status before population data were obtained (Methods) [registered: `results/s8086/tables/law_denominators.csv`; `results/s8086/tables/law_candidates.csv`; `results/s8086/S2.md`].

Exploratory, post hoc follow-ups address both the missing pusher and what the frozen extractor missed. No full `B9 51` tile in either phase, nor any alternating run of at least four bytes, was observed in 11,700 scheduled top-three sample records or 2,000,000 checkpoint cells. The archive therefore suggests a bottleneck in arrival or establishment, although its historical cause is not identifiable. Planted-pusher encounters show phase- and role-dependent vulnerability, not a measured population growth rate [exploratory: `results/s8086/exploratory/motif_scan.csv.gz`; `results/s8086/exploratory/encounter_fates.csv`; `results/s8086/EXPLORATORY.md`].

The dominant first-copier family instead evolved the same regenerating overlap geometry as the Z80, on a different instruction set. Ten of the 14 first copiers tile `5F A4 x y` (Fig. 5d). Post hoc tracing identifies `POP DI; MOVSB; branch`: the stack read initializes a displacement d = 4, and subsequent copies propagate the first four parental bytes. With this four-byte footprint fixed, free payload is zero, yet alternative branch bytes are transmitted through eight passages. Zero free payload is again not zero heritable variation [exploratory: `results/s8086/exploratory/first_family.csv`; `results/s8086/exploratory/first_family_copies.csv`; `results/s8086/exploratory/family_transmission.csv`; `results/s8086/exploratory/family_serial.csv.gz`].

Extending the dependency verifier post hoc admitted 600 genotype–world candidates: 590 full-tape d = 16 copiers and 10 d = 4 regenerators. Predictions were committed before assays on a purposive panel of 32 genotypes from 20 worlds, and every selected case agreed, with no mismatch in 2,224,640 encounters. The full-tape cases preserve arbitrary tested payload under held footprints; they are not smaller-seed regenerators. These results support transfer of the geometric explanation while leaving registered L1 and L2 untested [exploratory: `results/s8086/exploratory/eligibility_counts.csv`; `results/s8086/exploratory/eligibility.csv`; `results/s8086/exploratory/law_panel_predictions.csv`; `results/s8086/exploratory/law_assay_denominators.csv`; `results/s8086/exploratory/law_assay_summary.csv`].

Closure and heredity are distinct

In a recurring family of spontaneous copiers, overlap determines which parental bytes reach descendants; an inherited displacement controls free payload, and chemistry shifts the prevalence of the resulting designs. The processor comparison sharpens this claim. The prospective 8086 route prediction failed despite an available exact pusher, whereas exploratory analysis recovered the same overlap geometry with exact post hoc agreement. Availability does not decide the route; the copying law transfers where its geometric and execution conditions are met. A successful explanation of transmission need not be a successful prediction of evolutionary arrival.

The results separate closure to partner information from broad transmission of parental variation. With the parent and encounter contract fixed, closure means that offspring no longer depend on the partner's initial contents. A qualifying regenerator can achieve closure while preserving few tested alternatives, among them working core alleles, whereas transmitters preserve broad classes of variants at specific sites. Zero partner inflow is our deliberately strict analogue of Woese's transition towards stable vertical descent; Woese did not require horizontal transfer to disappear [6]. Confinement alone does not establish this closure, and the operational criterion is neither a necessary condition for Darwinian evolution nor a reconstruction of the historical cellular transition. Within these limits the Darwinian-threshold framing has content: stable, genealogically continuous reproduction can coexist with limited transmission of the variation on which selection could act.

The same distinction matters wherever replication is studied without genes. Recovery of a sequence or composition could reflect transmission, reconstruction or recapture from the environment. Following an introduced difference separately from the survival of its carriers, and identifying the source of recovered material, distinguishes these possibilities. We do not claim that chemistry uses a displacement byte. Ours is an inspectable case in which it is exactly decidable which alternatives reproduction preserves and which its own operation removes.

The remaining limits are substantive. Exact registered support comes from edited Z80 descendants and the eligible evolved 6502 panel; the 8086 follow-ups are exploratory and require a new prospective test with adequate extraction coverage. Finite joint challenges do not enumerate every possible payload, and neither useful payload function nor selection for increased capacity has been demonstrated: to establish adaptive use, an inherited difference must alter performance in a specified environment. Material that is both copied and interpreted, as in von Neumann's description, offers one route; whether such use evolves in these soups without an imposed task remains open [3]. Zilion makes these questions testable across explicitly specified processor contracts: not only whether a program reproduces, but which differences its reproduction preserves.

Figure legends

Colours are defined separately in each panel. Byte-grid entries are hexadecimal. Detailed protocols and numerical provenance are in Methods and Supplementary Notes 4–5.

Methods

Execution contracts

Zilion separates instruction cores, shared encounter interfaces, execution engines and machine manifests. Research code implements population pairing, mutation and scientific classifications. Manifests bind semantic version, compatibility, chip variant, generated shader and host identities to reset, address mapping, budget, traps and observations. Historical Algocell populations and assays retain their actual kernels; fresh reruns with corrected cores are reported separately. Architecture and machine-specific contracts are archived separately from experiment results [algocell_exp/machines/; /Users/neo/repos/zilion/docs/ARCHITECTURE.md].

An encounter places focal tape A and partner tape B on a ring, resets the processor, enters A, executes its budget and writes both halves back. Architectural wrapping precedes ring reduction; a wrap need not advance to the neighbouring ring byte for arbitrary ring sizes. The historical Z80 resets registers and flags to zero except SP, set to the largest architectural address aliasing the partner's last byte. That pointer convention does not itself specify the first written byte. Fetch maps include prefixes, operands and displacements, not merely instruction starts. Repeating block moves consume budget per repetition; I/O reads return zero, writes have no external effect and no external interrupts occur. A fetched zero opcode executes normally in benign chemistry; in lethal chemistry, a fetched zero opcode halts. The 8080 subset is an ablation within this core, not independent hardware validation [results/emulator/RING_DIFFTEST.md; algocell_exp/shader/z80_test_L16.wgsl; algocell_exp/shader/z80_test_lethal_L16.wgsl; results/review_r2/RUN_TABLE.md].

The NMOS 6502 starts with clear accumulator/index registers, status `20` hexadecimal and a first push aligned to the partner's last byte. Zero-page and stack-page architectural accesses map into the ring. Faithful BRK vectoring and halt-before-BRK are separate contracts; JAM halts, without external IRQ/NMI. BFF's published contract halts on unmatched brackets and pointer run-off; its literal, bandwidth, pointer and halting variants remain separate [results/s6502/CORRECTION.md; algocell_exp/machines/m6502_pair_maps_L16.wgsl; results/bff/runs.csv].

Coherent-fetch 8086

S2 used the Zilion 8086 core frozen at public commit `05bd6fa2219c90b34574189864631a42ef80ea66` (recorded in the frozen manifests under its byte-identical pre-publication identity `39a60429a9b16c77cd845f6087062c4e2cc12c53` [algocell_exp/machines/ZILION_PUBLIC_COMMITS.md]), compatibility `0.3`, with `prefetch=coherent` and `hidden_latch=zero_ea`; package status was experimental. Consumed instruction bytes come from current memory. There is no prefetch queue or bus-timing model. General registers, segments and entry IP reset to zero except SP = 1; FLAGS is `F002`, with DF and TF clear. A word push decrements SP by 2 modulo 65,536. At the registered ring sizes, the first word push writes low then high bytes at ring positions $P-1$ and 0. These results use this reset rather than the exploratory zero-SP inventory [algocell_exp/machines/worlds_8086.json; algocell_exp/machines/PROVENANCE_8086.md; results/s8086/prelaunch/freeze.json].

Segment translation and architectural wrapping precede ring mapping; each word byte wraps its offset independently. A REP iteration consumes an execution step and retains decoded operation/source segment between continuations; zero count copies nothing. Register LEA/LES/LDS forms trap. Undefined far-register transfers use the declared fresh zero-EA

hidden-state abstraction. Interrupt vectors alias ring memory, external interrupts are absent, input supplies all-one bytes, output is discarded, HLT halts and WAIT without a coprocessor does nothing. These policies matter to evolving programs; coherent fetch is the research contract, not a claim to reproduce hardware-prefetch ecology [algocell_exp/machines/worlds_8086.json; algocell_exp/machines/i8086_pair_maps_L16.wgsl; results/s8086/prelaunch/freeze.json; /Users/neo/repos/zilion/docs/I8086.md].

Worlds and detection

Emergence worlds start with pseudorandom byte tapes generated from the declared seeds; packed padding bytes are excluded from the tapes. Competition worlds instead start from the specified spatial mixtures of partial-copy and full-tape designs [algocell_exp/soup.py; offset_switch.py].

The shared soup contains 20,000 cells on a 160 × 125 lattice with reflecting cardinal-neighbour boundaries. Each update draws 8,192 ordered pair slots; collision claims determine active encounters, so draws are not executed-pair counts. The baseline encounter budget is 128 execution steps; S1 arm C uses 512. At noise exponent 4, mutation makes 512 uniform byte-replacement attempts per update. Packed-word collisions can discard mutation writes, and replacement may leave a byte unchanged. Applied mutation counts are unmeasured. Pairing races also mean seeds do not guarantee identical whole-world replay [algocell_exp/soup.py; s8086.py; results/review_r2/RUN_TABLE.md; results/s8086/prelaunch/registration_basis.json; results/s8086/prelaunch/freeze.json].

S1 comprised four arms of 20 worlds, each run for 10,000,000 updates: A, L = 16/budget 128/faithful BRK; B, L = 32/128/faithful BRK; C, L = 16/512/faithful BRK; D, L = 16/128/halt-before-BRK. Seeds 65001–65020 were reused across arms, giving paired initial conditions where lengths matched. Samples recorded the top-three tapes every 50 updates to 2,000, every 250 to 20,000, every 2,500 to 1,000,000 and every 25,000 thereafter. The 6502 retention study (registered experiment C2) used budget 128 even for candidates originating in arm C [REVISION_PREREG.md, S1; results/s6502/CORRECTION.md; results/evolved6502/coverage.csv]. Supplementary Table 1 gives cohort-specific selection rules, seeds, schedules and horizons.

S2's sole population arm comprised 20 L16 worlds, seeds 8086001–8086020, through 3,000,000 updates. L32 population arm B was withdrawn; pointer randomization was omitted because it lacked a controlled census. L32 remains fixed-input census/control context only. Each world retained 585 scheduled records, including initialization, and complete checkpoints at 0, 20,000, 100,000, 1,000,000 and 3,000,000 updates. Sampling intervals widen according to the frozen schedule; they are not adjusted after detection [s8086.py; manuscript/figures/out/v8/data/arrival_worlds.csv; results/s8086/tables/worlds.csv; results/s8086/tables/download_hashes.csv; results/s8086/prelaunch/freeze.json].

For S2, detection is the first scheduled top-three exact tape at share at least 0.005 that passes culture, choosing greatest share then lexicographic order. Final random-cell sampling does not retroactively change detection. Times retain preceding-sample intervals. Complete nondetections are right-censored at horizon; missing or corrupt records are incomplete, not biological negatives. Kaplan–Meier summaries use worlds as replicates. Cells, roles, alternatives and descendants are nested observations; none enlarges the number of independently evolved worlds [score8086.py; results/s8086/tables/worlds.csv; results/s8086/tables/kaplan_meier.csv; results/s6502/v2/arm_comparison.csv; REVISION_PREREG.md, S2].

Culture, carriage and inflow

Culture scores normalized gain in best-cyclic-shift similarity to the original parent, `(after−before)/(1−before)`, averaged over informative partners with before-match below 0.75. First offspring reproduce against fresh partners. The generation-two score includes all first-generation outputs, including those that failed to become close copies; it is not conditioned on a successful first copy [algocell_exp/assay.py]. Historical "heritable" means generation-two score at least 0.3, not exact inheritance. S2 used 64 uniform partners per generation, including final-cell cultures. NaN is unassessable, not zero. S2's similarity and faithful-offspring fields are reported but do not add undeclared detection gates [manuscript/figures/out/v8/data/methods.csv; results/s8086/tables/copier_tapes.csv; results/s8086/prelaunch/registration_basis.json].

S1 specified 64 culture partners for heritability, whereas its cited frozen final-cell estimator used 32; top-tape cultures used 64. The trace text specified soup partners, whereas the frozen implementation used 16 uniform-byte partners. Corrected frozen32 results are primary. Complete 64-partner final-cell reclassification and first-tape soup-snapshot tracing are separate sensitivity analyses. Neither sensitivity changed the reported first-tape classifications or final-cell counts [results/s6502/CORRECTION.md; REVISION_PREREG.md, S1].

Single-site scans ask whether a reproductively competent mutant transmits its changed byte into at least half of qualifying aligned offspring. A site needs at least half its tested identifiable values to pass. Population assays classify confined cells with at most two scored sites as regenerators, three or four as intermediate and at least five as transmitters. These thresholded classes do not measure exact capacity. "Unexecuted" means absent from instruction-stream fetches, including operands; it does not exclude data use [mutscan.py; results/ladder/LADDER.md; results/population/classes_snapshots.csv; results/crystal/cells.csv].

At each transfer, an offspring qualifies only if at least 75% of its positions match its immediate parent under the best cyclic shift. A serial lineage survives through generation g only if every transfer qualifies. Branching populations retain qualifying offspring, without carrying parents forward; when more than 32 qualify, 32 are sampled uniformly. A branching population survives if at least one member remains [serial_retention.py; branching_transfer.py; REVISION_PREREG.md, X1].

Historical serial assays follow the centred five-byte mutant window, permitting cyclic alignment and correcting carriage against surviving unmutated controls. Windows already present in wild type are unidentifiable and excluded. The panel uses 16 mutant lineages, 256 controls and eight transfers; alternatives are exhaustive for short tapes and sampled for longer ones. Branching X1 instead gives each survivor two partners, caps populations at 32 and uses four populations for each of 32 alternatives/site plus 64 controls. A population carries when at least half its members carry the marker; site scoring requires the registered 0.5 excess over control carriage among survivors. Fate fractions in Fig. 2a use all original starts. The soup-partner arm pools open-phase worlds rather than the focal neighbourhood; pre-existing matches and reacquisition can inflate carriage [results/serial_retention/sr_summary.csv; results/serial_retention/REPORT.md; results/branching/branching_summary.csv; results/branching/BRANCHING.md].

For deterministic offspring Y = F(x,E), fixing parent and contract gives I(E;Y|X = x) = H(Y|X = x). Exact copying implies zero inflow, but zero inflow does not imply preservation of parental variation. Sampled zero entropy is not partner-universal closure. The historical entropy panel has 256 outputs and an eight-bit sample ceiling, with downward undersampling bias. Fetch confinement, original-partner access proxies, output variation and parental transmission therefore

remain separate [theorem: manuscript/SI_v8.md, Supplementary Note 2; results/biology/individuality/per_replicator.csv].

Competition endpoints

Registered competitions seed partial-copy R and full-tape T designs in equal mixtures or reciprocal rare invasions and retain mutation-on/off and benign/lethal arms. The designs differ in payload background as well as displacement, so environmental sorting does not isolate selection on capacity. At L32, classification requires `5e ed b0` at bytes 1–3 and byte 0 in R = {04,44,84,08,48,88,C8,50,90} or T = {60,A0}, all hexadecimal; other values remain unclassified. This census-defined classification differs from the $\leq 2/\geq 5$ site-score classes of random-population assays [offset_switch.py; results/offset/OFFSET_CENSUS.md; REVISION_PREREG.md, E]. Fig. 4 shows L32 mutation-on runs through 300,000 updates: ten equal-start worlds and five per invasion direction for each chemistry/cohort. Relative share is T/(R+T); broader core-motif prevalence uses all cells. Undefined ratios when both classes disappear are omitted only from ratio summaries, retaining absolute loss and zero tail diversity. Trajectories from the historical implementation and the corrected Z80 core (core C1) are summarized separately at world level [results/offset/offset_switch.csv; results/tier2/scored/E/offset_switch.csv; manuscript/figures/out/v8/data/competition_endpoints.csv].

The lethality dial varies the probability that a fetched zero opcode halts, with nondetections censored at its horizon. Under the corrected core, Kaplan–Meier medians of first-confinement time at p = 0, 0.01, 0.03, 0.1, 0.3 and 1 were 32,500, 42,500, 60,000, 80,000, 212,500 and 115,000 updates. This series has one reversal, so DZ2's monotone component was met; historically it was not met (two reversals). The registered endpoint is the first sample containing a heritable top-three tape at $\geq 0.5\%$ of cells with no partner instruction-stream fetch in 16 of 16 encounters. It missed confined transmitter populations in two historical worlds, so the test is biased against confinement at high p [results/tier2/T2.md, DZ2; REVISION_PREREG.md, DZ2]. The tested toxin, backup-core and mutation-only effects did not meet their registered explanatory criteria, and exploratory intruder assays identify a lead rather than a completed mechanism [results/tier2/T2_DZ.csv; results/toxin/CAUSAL.md; results/n2/BACKUP.md; results/offset/OFFSET_FLOW.md; results/offset/INTRUDER.md].

Forward-overlap theorem

Theorem. On a ring of P bytes, perform ordered writes M[s+d+k] ← M[s+k], modulo P, with $0 < d < P$ and no interfering operation. Before k = d, reads obtain original source bytes; subsequently each source was last written d assignments earlier. Induction gives ancestral coordinate s+(k mod d). With d dividing P and coverage at least P−d, the protected seed tiles the ring; with P = 2L and d dividing L, the offspring has the required phase. If a fixed constrained footprint C lies in the protected seed, independent free payload is r = d−|C|. This describes a product family of 256^r sequences, or 8r ideal noiseless bits, not ecological neutrality or measured Shannon capacity [theorem: manuscript/SI_v8.md, Supplementary Note 1; results/capacity/geometry_traces.csv.gz].

For the Z80 `XX 5e ed b0` specialization, $L \leq 64$, d = XX mod 2L, $4 \leq d \leq L$ and d divides L. The first instruction must pass control to address 1, preserve the needed source/base state, write only outside the protected prefix and leave HL ≡ 0 and 256D ≡ 0 modulo 2L, with BC zero or at least 126. Under the declared initialization, 126 block moves remain; the immediate `1e NN ed b0`

form leaves 127. Architectural address increments must preserve displacement. Core preservation keeps execution in the copy when overwritten instructions are revisited. Free coordinates then survive jointly and serially; reverse `LDDR` requires another derivation [theorem: `manuscript/SI_v8.md`, Supplementary Note 1, Z80 specialization; `results/capacity/candidate_eligibility.csv`].

Working core alleles lie outside fixed-footprint payload. Visible sequence period is not displacement. Motif/divisibility screening supplies candidates, not every proof hypothesis. The exploratory population comparison retains its whole-tape gate deficit and additional fetched-site scores rather than redefining the theorem around measured sites [`results/crystal/cell_exceptions.csv`; `results/crystal/offset_examples.csv`; `results/offset/offset_cells.csv`].

Provenance certificates

A sufficient certificate fixes the admissible parent family, reset, partners, budget, descendant boundary and alignment. Original parent/partner coordinates receive symbolic labels; each pure copy inherits its source's current label in ordered execution. The proof must establish complete offspring provenance, noninterference under jointly arbitrary admitted payload assignments, recoverability of those assignments from the offspring, viable offspring, preserved entry machinery and membership in the same family after reset. This makes serial induction nonvacuous. One trace or matching byte values alone cannot prove the family claim. Arithmetic encodings and value-dependent programs may transmit information outside this sufficient pure-copy criterion [theorem: `manuscript/SI_v8.md`, Supplementary Note 2; `results/capacity/candidate_eligibility.csv`; `results/evolved6502/eligibility.csv`; `results/s8086/prelaunch/ASTRA_PRELAUNCH_REVIEW.md`].

Amended S2 L2 requires the actual sampled baseline to be d-periodic at physical entry zero, with d dividing L, full partner coverage and no original-partner input or interference. It permits original roots $R = \{s,\ldots,s+d-1\}$ inside A. For physical offspring position q, predict $a(q) = s+((L+q-s) \bmod d)$. Independently extract raw core K, retaining the bound $c = 1+\max(K) \leq d$, and define source core $C_R = \{a(k):k\in K\}$, free set $F = R\backslash C_R$ and erased set $E = A\backslash R$. Hold both K and C_R. The exact free count is $d-|C_R|$, not necessarily $d-c$. Mutable erasures are $E\backslash K$; erased setup coordinates $E\cap K$ remain held because their values may govern execution even though their original ancestry disappears. No dependency is reduced modulo d to hide a setup requirement. Source phase and physical entry are distinct; compare physical B directly, without favourable rotations. The certified map composes with itself across transfers [theorem and registered amendment: `REVISION_PREREG.md`, S2 B2; `results/s8086/prelaunch/ASTRA_PRELAUNCH_REVIEW.md`; `results/census8086/LAW_CONTRACT.md`].

Z80 displacement interventions

The Z80 displacement study (registered experiment C1) was registered at `213a653`; eligibility was frozen at `4855ac3`. Its 22 archived parents include fixed representatives and deterministically ordered eligible archive rows, not independent worlds. All 101 settings qualified before retention. The default archived Z80 executor, entry-phase core, dividing displacement, ordered copying, coverage and noninterference were checked against zero plus 16 frozen partners. Initial tails need not be periodic: regeneration is the predicted offspring, not extinction [`results/capacity/parents.csv`; `results/capacity/edit_eligibility.csv`; `results/capacity/partners.csv`; `REVISION_PREREG.md`, C1].

Forward passages followed 16 lineages through eight generations; reciprocal passages restored the original displacement byte in actual descendants and repeated transfer. Complete core,

displacement, offspring and ancestry are retained in 25,856 serial rows. Direct assays used actual generation-eight descendants, testing all 255 alternatives at every free and erased coordinate plus 64 joint assignments against all 17 partners. Deterministic duplicate descendants share execution while retaining lineage identities. All 1,067,608 direct cases and all serial observations matched. These are intervention cases, not encounter counts. Joint sampling is finite; the unrestricted joint claim is conditional mathematics [results/capacity/serial.csv.gz; results/capacity/capacity_cases.csv.gz; results/capacity/CAPACITY.md; results/capacity/bursts.csv].

Evolved 6502 retention

The 6502 retention study shares registration `213a653`. The original extractor and 714-role ledger were frozen at `943aeee`; unchanged extraction extended to 942 roles at `657eb7e` before interventions on newly recovered top-three roles. The descriptive-copier inclusion clarification at `f57ad24` preceded new assays. Corrected S1 frozen32 primary calls determine the descriptive final-copier panel; every registered sampling role remains in the geometry ledger. Sensitivity analyses reclassify the same sampled cells. All raw logs matched their frozen input hashes, with no substitute tapes [results/evolved6502/coverage.csv; results/evolved6502/inputs.csv; results/evolved6502/RETENTION.md].

The complete panel contains 62 first detections, 640 final random draws and 240 final top-three roles from 80 worlds. It has 861 distinct tapes and 42 eligible roles. Footprint c counts constrained coordinates, not necessarily a contiguous prefix. Ordered provenance admits pure-copy channels and rejects or leaves unresolved unknown arithmetic, reverse/nonconstant motion, partner inflow, interference and uncertified control. Using each world's frozen 6502 machine at budget 128, every admitted role received all 255 alternatives at each free/erased coordinate and 64 joint assignments against zero plus 16 uniform partners, followed through eight fresh-partner generations. Complete physical offspring and source labels are scored without similarity gates or cyclic realignment. There were 91,428 intervention cases and 731,424 serial observations, without mismatch. A separate descriptive serial scan uses marker thresholds on ineligible copiers: two first and two final roles per arm, 32 alternatives/site, four mutant lineages and 64 controls. It neither tests the exact law nor supplies a registered directional prediction [evolved6502_retention.py; capacity_common.py; results/evolved6502/panel.csv; results/evolved6502/eligibility.csv; results/evolved6502/geometry.csv; results/evolved6502/law_verdict.csv; results/evolved6502/descriptive_panel.csv; results/evolved6502/exact_cases.csv.gz; results/evolved6502/exact_serial.csv.gz; results/evolved6502/bursts.csv].

Literal-channel results

Theorem. A repeated pair with distinct one-byte load/push opcodes loads w literal bytes into one register and pushes a fixed permutation without changing it, using a separate stack pointer that moves by $-$w per push. Under contiguous noninterfering writes and the prescribed executed stream, input/output periods are w+2 and w. The Fine–Wilf periodicity theorem [15] requires N $\geq$ 2w+2$-$gcd(2,w) for finite-window equality. Odd widths then force identical opcodes, a contradiction; even widths permit compatible alternating constructions. Descending stacks reverse block order in ascending memory, not internal byte order. Placement, bootstrap and coverage remain separate [theorem: results/literal_channel/CHECK.md; results/literal_channel/pair_layouts.csv; results/literal_channel/finite_bound_counterexamples.json].

Theorem. For width-one periodic mixtures through a shared register, a minimal decoder cycle visits at most d residues of primitive input period d. If it contains loads and pushes, emitted length n_P satisfies $0 < n_P < t \leq d$, incompatible with d dividing n_P for an equal infinite stream. Multiple opcode identities do not change this. The degenerate all-push writer initialized with an opcode-valued register is excluded. Width-three mixtures provide counterexamples to extending parity beyond repeated pairs. A finite width-one factor witness exists but fails archived zero-partner bootstrap. Only three of the five nontrivial Z80 period-two writers are literal; the others are memory-fed. None of these statements proves an evolutionary arrival order or excludes every finite 6502 copier [theorem/construction: `results/literal_channel/verification.csv`; `results/literal_channel/finite_search.csv`; `results/literal_channel/6502_finite_harness.json`; `results/literal_channel/census_audit.csv`].

Causal census classification

The validated 8086 census evaluated all 256 period-one and 65,536 period-two words at L16/L32 and budgets 128/512 against zero plus 16 frozen partners. Constants and reversed phases were retained, with no unevaluated words. Mechanism descriptions use the causal classification below. Its audit covers 1,246 settings and 21,182 encounters; 1,247 labels changed without changing outputs or matches [`results/census8086/table_main.csv`; `results/census8086/table_hits.csv`; `results/census8086/descriptive_correction_counts.csv`; `results/census8086/DESCRIPTIVE_CORRECTION_2026-10-12.md`].

The classifier examines execution through the last partner write. Every claimed store, including neutral read-modify-write, must pass isolated `55` and `AA` source interventions with matching ordered addresses through that write and the expected source-dependent value. Later divergence is allowed. Literal provenance, paired memory copying and confined copy loops are separate; loops require repeated full instruction-start addresses, not merely mapped ring positions. Mixed, unsupported, failed, vector-dependent or overflowing cases remain "other". Trace memory and original records must agree with uninstrumented execution. Full-encounter partner fetch and pre-write access proxies are separate from this causal classification [`results/census8086/MECHANISM_CONTRACT.md`; `results/census8086/DESCRIPTIVE_CORRECTION_2026-10-12.md`; `results/s8086/tables/judged_encounters.csv.gz`].

Repeated classifier corrections undermined the rationale for R2's numerical predictions, so R2 occurrence and persistence were withdrawn prospectively before population data; they remain descriptive. R0/R1 use independent culture/fetch endpoints. The `07 A5` witness remains causally memory-fed in all 68 audited contexts and open in all 64 random-partner contexts. At budget 128, `B9 51` bootstraps against zero and copies exactly into 13/16 random partners at L16 and 6/16 at L32, so it is partner-dependent; `FF 30` and `AD 50` are not exact zero-partner bootstrappers under this reset. Available mechanisms are not establishment probabilities [`results/s8086/tables/census_07a5.csv`; `results/census8086/table_hits.csv`; `results/census8086/words_L16_k2_b128.csv`; `results/census8086/words_L32_k2_b128.csv`; `REVISION_PREREG.md`, prospective R2 withdrawal].

Registered 8086 tests

R0 requires at least 10 detections among 20 worlds. Given that sufficiency, R1 requires at least ceil(0.8D) open first copiers; the kill rule triggers at ceil(0.25D) confined first copiers. Open means partner instruction-stream fetch in at least 8/16 complete traces; confined means none in all 16.

Intermediate histories remain in D. Observed D = 14, open = 2 and confined = 12 give R0 pass, R1 fail and the route kill. R2 records occurrence in 1/20 worlds and persistence in 0/20, without confirmatory cutoffs. Confinement timing has median 975,000 updates, with eight censored worlds; it is not a causal-closure or loop endpoint [results/s8086/tables/worlds.csv; results/s8086/tables/persistence.csv; results/s8086/tables/kaplan_meier.csv; results/s8086/S2.md].

Law candidates comprise first detection, first confinement, final top-three tapes and 64 uniformly sampled final cells per world, preserving cell multiplicity. Eligibility was committed at `8971d0` before evolved interventions. L1 admits only the verified six-byte MOV-DI/DEC-CX/REP-MOVSB-or-MOVSW family; L2 uses the amended certificate above, with separate with/without-L1 reporting and per-family caps. Of 1,337 distinct genotype-world candidates, 779 were culture-positive. L1 excluded all; L2 excluded 55 for original-partner reads and left 724 unresolved, including 708 at POP ES. No cap excluded a case, and no evolved law intervention occurred. L1 is untested; L2 is untested with support suspended. The frozen reporter's `incomplete/invalid` extraction label does not invalidate the completed route cohort [results/s8086/tables/law_candidates.csv; results/s8086/tables/law_denominators.csv; results/s8086/tables/law_worlds.csv; results/s8086/S2.md].

L1 requires the sampled baseline to be d-periodic at physical entry zero, $6 \leq d \leq L$ with d dividing L and even d for MOVSW, reset source/segments, sufficient ordered forward coverage, preserved core and no interfering write or original-partner read. Each family could select at most 16 genotype-world cases per world, taking eligible first/closed/final roles before keyed random-cell candidates. The registered design tests all 256 values at free and mutable erased coordinates, 256 counterbalanced plus 256 seeded joint assignments, all 17 census partners and eight serial generations for singles/joints. A reproducible mismatch remaining independently eligible fails the application; unresolved certificates suspend support. These planned evolved assay denominators remained empty [results/census8086/LAW_CONTRACT.md; results/s8086/tables/law_outcomes.json; REVISION_PREREG.md, S2].

Constructed controls are separate evidence. The preceding census qualified 340 geometry/partner/budget rows; its six-byte construction gives 2/10 free coordinates at L16 and 2/10/26 at L32 under the admissible displacements [construction: results/census8086/table_law.csv]. All 10 byte/word REP controls at the registered budget completed with zero mismatches in 1,185,792 encounters, including fixed-input L32 geometries. They test complete physical offspring, ancestry and serial reset closure; they supply no evolved support [construction: results/s8086/tables/constructed_controls.csv; results/s8086/tables/control_hash_verification.csv].

Restart and scoring continuity

Publication failure required a restart from initialization before biological outputs were read; no trajectory was spliced. Scoring recovery did not rerun completed worlds or change scientific definitions (Supplementary Note 6) [results/s8086/prelaunch/S2_OPERATIONAL_AMENDMENT_2026-10-12.md; results/s8086/prelaunch/S2_VOLUME_PARALLEL_AMENDMENT_2026-10-12.md; results/s8086/prelaunch/S2_SCORING_ONWARD_LOCK_AMENDMENT_2026-10-12.md; results/s8086/provenance/run_provenance.json].

Exploratory 8086 analyses

Post hoc scans covered both pusher phases in all scheduled top-three records and complete checkpoints. Encounter assays planted tiles in focal and receiving roles against sampled soup and matched uniform partners. Absence of observed seeds and receiving-role vulnerability motivate an arrival/establishment interpretation, without identifying historical cause, unseen births or population growth. Founder-byte transmission varied homologous positions across the repeated tile; law erasure tests instead altered single physical coordinates [exploratory: `results/s8086/exploratory/motif_scan.csv.gz`; `results/s8086/exploratory/encounter_fates.csv`; `results/s8086/exploratory/family_transmission.csv`].

A separate extended dependency verifier re-extracted all 779 culture-positive candidates, admitting 600 and leaving 72 excluded and 107 unresolved. The purposive 32-genotype panel represented 20 worlds; predictions were committed at `5bbc9d7` before assays, but extraction and selection remain post hoc. The frozen assay was reused with the extended verifier and 64 independent joint assignments in place of 512. All-value single/mutable-erasure substitutions, 17 census partners and eight serial transfers yielded 2,224,640 encounters without mismatch. For d = 4, free payload is empty and joint assignments repeat baseline; for d = 16, erased coordinates are empty. CPU audits verified hashes, source maps and denominators. No new population worlds were run, and the registered verifier and verdicts were unchanged [exploratory: `results/s8086/exploratory/eligibility_counts.csv`; `results/s8086/exploratory/eligibility.csv`; `results/s8086/exploratory/law_panel_predictions.csv`; `results/s8086/exploratory/law_assay_denominators.csv`; `results/s8086/exploratory/verification.csv`; `results/s8086/EXPLORATORY.md`].

Validation and chronology

Validation distinguishes instruction, ring-encounter, isolated-reproduction and ecological levels. Z80/6502 vectors are emulator-generated; 8088 vectors are hardware-recorded. The latter give 3,126,917 raw passes among 3,137,000 cases, with 10,083 expected hidden-latch divergences and none unexpected; queue state and timing are outside scope. A same-core CPU implementation is not an independent oracle, and deterministic encounter replay does not establish deterministic soup replay [`manuscript/figures/out/v8/data/validation_fractions.csv`; `results/emulator/RING_DIFFTEST.md`; `algocell_exp/machines/PROVENANCE_8086.md`].

I1 compared historical compatibility core C0 with the corrected core and the vendor-specific Zilog-NMOS Z80 core (core C2). It flagged 47/690 and 114/690 conditions respectively, with 407/417 isolated combinations identical in each comparison; the all-unaffected criterion failed. Omitted named Stage-E cases were completed locally with the sealed engine. Stochastic assays use case-index streams, not archived encounter streams. I1-T2 retained 19/22 claim verdicts but changed four of 36 components, including historical DZ2/DZ3 failures becoming passes and Stage-L L2 becoming not applicable at L16 and not met at L20, while its joint kill still fires. Preserved verdicts include the failed E1 and E2 abundance predictions (Supplementary Note 5). The Stage-L dispatcher's missing-snapshot alerts were retained; frozen scorers used complete required records. Fresh nondeterministic worlds do not isolate a causal core-repair effect [`results/impact/decisions.csv`; `results/impact/assay_identity.csv`; `results/impact/supplement_manifest.json`; `results/tier2/T2_CLAIMS.csv`; `results/tier2/T2_COMPONENTS.csv`; `results/tier2/stageL_diagnosis.json`].

The chronology in Supplementary Information distinguishes discovery from confirmation. X1's carriage prediction failed; the first X3 archive existed before registration, whereas its replication was prospective; N6's failed component remains failed. The validated 6502 census preceded S1, while the literal theorem is retrospective. S1 correction retains unsupported P1, untested P3 and the registered-versus-executed partner-count/source discrepancies. The 6502 retention study's later exact-law pass does not rescore S1. Crystal, ladder and the 8086 follow-ups are exploratory. S2's final scientific amendments preceded population data; subsequent restart and scoring recovery preserved the scientific definitions [REVISION_PREREG.md; results/lod_n6/N6_SCORE.md; results/s6502/CORRECTION.md; results/evolved6502/coverage.csv; results/s8086/prelaunch/ASTRA_RECHECK3.md; results/s8086/exploratory/publication_archive.csv].

Statistical analysis

Independent evolutionary replicates are worlds; sampled cells, roles, variants, partners and serial observations are nested within them. Deterministic intervention mismatches are enumerated, not treated as independent-world trials. Detection summaries use scheduled event times and Kaplan–Meier right censoring at the declared horizon. Competition summaries use world-level medians; ratios with R+T = 0 are undefined and excluded only from ratio summaries. Sample sizes and exclusions are given for each cohort. The benign corrected-core T-into-R subgroup has four defined ratios from five worlds. Supplementary Notes 5 and 7 describe the applicable confidence-bound and resampling procedures; exact law checks do not receive independent-world confidence intervals [manuscript/figures/out/v8/data/methods.csv; manuscript/figures/out/v8/data/competition_endpoints.csv; results/impact/decisions.csv].

Source data and figure provenance

Generated source tables provide the figure coordinates and empirical counts. Supplementary Note 5 defines the cohorts, measurement units, exclusions and display selections; Supplementary Table 15 identifies the figure source data and provenance manifests.

AI-assisted work

[AUTHOR TO CONFIRM: AI-use disclosure.] Claude (Anthropic) and GPT-based Codex agents (OpenAI) assisted with code implementation, analysis scripts, adversarial review, figure production and drafting. The author directed the study, checked the results against committed generated tables and takes responsibility for the content.

Supplementary Information and displays

The accompanying `manuscript/SI_v8.md` contains the full overlap-copy theorem and Z80 specialization (Supplementary Note 1), the provenance certificate and inflow distinction (Supplementary Note 2), scoped literal-channel theorems and constructions (Supplementary Note 3), machine-specific copying-law conditions (Supplementary Note 4), cohort and assay definitions with detailed figure provenance (Supplementary Note 5), registration chronology and restart exclusions (Supplementary Note 6), I1/I1-T2 impact results (Supplementary Note 7), the causal census correction (Supplementary Note 8) and the exploratory 8086 analyses (Supplementary Note

9). Supplementary Tables 1–3 identify cohorts, registration chronology and complete verdict ledgers; the remaining numbered tables identify generated supporting records. The Extended Data map lists each display, its `ed8_*.pdf` target and its supporting sources [manuscript/SI_v8.md].

Data availability

Generated reports and figure data are present in this repository under `experiments/results/` and `experiments/manuscript/figures/out/v8/`. Each result family retains generated tables, input identities, protocol references and status. Result-path brackets identify the supporting records. S2 additionally retains frozen reporter outputs, source and download inventories, failed-attempt provenance and eligibility release [results/s8086/provenance/artifact_manifest.json; results/s8086/provenance/verification.json; results/s8086/exploratory/publication_archive.csv].

Some full raw logs and snapshots are local or externally stored rather than committed; their expected paths and hashes are recorded in experiment-specific manifests. Hashes and local absolute paths do not supply review access. [AUTHOR TO SUPPLY: data accession/DOI and accessible archive containing required raw records, source-data tables, preregistrations, amendments, the standalone cross-machine encounter benchmark package with its generated table and a release-wide checksum manifest, with durable retrieval locations and any access restrictions.]

Figs 1–5 have PDF and PNG exports in `manuscript/figures/out/v8/`. Fig. 3's registered 8086 law test remains untested; the separate exploratory assays are not plotted. The Supplementary Information maps Extended Data displays to their files and sources [manuscript/figures/out/v8/FIGURES_V8.md; manuscript/SI_v8.md].

Code availability

The current repository contains historical Algocell research execution and analysis code; absolute Zilion paths identify external source locations, not public releases. Capacity and retention reports are generated by `capacity_assays.py` and `evolved6502_retention.py`. The registered S2 presentation is regenerated by `results/s8086/provenance/generate_report.py`; exploratory analysis and its hash audit use `exploratory8086.py`, `exploratory8086_law.py` and `exploratory8086_verify.py`. Pre-assay prediction identities establish the ordering of registration, eligibility and tests. Report regeneration is distinct from running new simulations [results/s8086/provenance/artifact_manifest.json; results/evolved6502/inputs.csv; results/capacity/CAPACITY.md].

Figure reproduction uses `manuscript/figures/prepare_v8.py`, `build_v8.py`, `legends_v8.py` and `verify_v8.py`; reproduction instructions are recorded in `manuscript/figures/out/v8/FIGURES_V8.md`. [AUTHOR TO SUPPLY: frozen Algocell and Zilion code releases and accession/DOIs, dependency versions, licence, hardware requirements and verified runnable instructions distinguishing report regeneration from new simulations.]

Author contributions

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[AUTHOR TO SUPPLY: competing interests declaration.]

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